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Effective Installation Solutions of Subsea Pipelines in Shallow and Ultra-Shallow Water

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Abstract

The installation of pipelines in shallow and ultra-shallow water environments presents unique challenges due to the dynamic and testing nature of these settings. This paper aims to provide a comprehensive analysis of the various methods employed in the installation of pipelines in such environments, highlighting their advantages, limitations, and suitability for different conditions.

The paper begins by outlining the key factors influencing pipeline installation in shallow waters, including hydrodynamic forces, seabed conditions, and the transitional interface between onshore and offshore zones. It then presents a case study from the northern Caspian Sea region, focusing on the installation of a pipeline through shallow water using traditional methods such as On-Bottom, Surface Tow and Float and Flood, Off-Bottom, and a Hybrid approach that integrates Surface Tow and On-Bottom methods. Each method is assessed against criteria including installation speed, vessel and equipment availability, constructability, and overall feasibility in light of project-specific constraints.

Due to operational requirements, the 32 in outside diameter (OD) concrete weight coated pipeline necessitated an internal mechanical corrosion-resistant alloy (CRA) lining. This lining imposed a limiting bending strain of 0.216%, which became a key constraint in evaluating installation methodologies, particularly those involving floating sections. Feasibility assessments and configuration options were guided by this strain limitation. Various string lengths were analysed during towing to balance maintaining strain levels within the CRA's allowable limits while maximising string lengths to reduce offshore activities, especially complex above-water tie-ins (AWTIs). This approach aimed to deliver a practical, installable solution aligned with project-specific constraints.

It is worth noting that loads induced during flooding operations and potential fatigue damage were not considered and fall outside the scope of this paper.

This paper synthesises contemporary knowledge and industry practices to inform engineers and project managers responsible for the planning and execution of pipeline projects in shallow and ultra-shallow water environments, particularly those involving complex shore approaches.

Introduction

Installing pipelines in ultra-shallow and shallow water environments presents a unique set of technical, environmental, and logistical challenges. These include highly dynamic hydrodynamic forces, environmental and ecological sensitivity, challenging terrain, limited accessibility, variable sea and weather conditions, and operational constraints. Combined, these factors contribute to increased installation complexity compared to deeper water projects, often necessitating hybrid solutions to effectively manage the onshore to offshore interface.

Each project is unique, presenting a distinct set of challenges that require innovative solutions. In some cases, environmentally sensitive areas restrict the use of intrusive burial techniques such as trenching and dredging, necessitating the adoption of trenchless installation methods. Shallow gradient beaches pose additional complexities, including extended intertidal zones. These conditions complicate construction timing, particularly in regions with large tidal ranges. Sediment transport is typically most active in the intertidal zone, where nearshore breaking waves are prevalent, further increasing the difficulty of trenching and backfilling operations.

Shallow water limits the use of large pipelay barges and high-bollard support vessels, which require deeper drafts for safe operation. This constraint often calls for shallow-draft barges, multicats, or modular pontoons. When conventional vessels or techniques are unfeasible, alternative installation methods such as Float and Flood, Off-Bottom, or On-Bottom are employed.

The interface between onshore and offshore dredging is typically defined by water depth and designated at Mean Low Water Springs. Onshore trenching, subject to ground conditions, is generally carried out using excavators with the use of a sheet-piled cofferdam (see Fig. 1). To extend onshore works and reduce the scope of offshore operations, which are typically more costly, a causeway can be constructed adjacent to the cofferdam (see Fig. 2). Offshore trenching requires a dedicated dredging spread and in shallow waters, Backhoe Dredgers and Cutter Suction Dredgers are common methods employed. The selection of the appropriate dredging vessel depends on seabed geology and equipment availability and should be made in close collaboration with specialist dredging contractors.

Careful consideration of installation loads and the resulting induced pipeline stresses is essential when selecting an appropriate installation methodology. These loads typically influence the choice of installation equipment such as marine tow vessel capacity and linear winch capacity, and can have a significant commercial impact on the overall project.

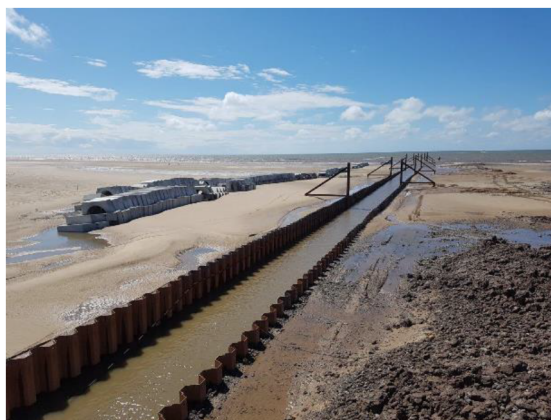


Figure 1—Photograph of Sheet Piled Cofferdam



Figure 2—Photograph of Sand Formed Causeway with Onshore / Offshore Interface

The stresses induced in a pipeline during installation are generally a combination of axial, bending, hoop, and shear components. These are typically assessed using analytical methods during the design phase. However, when a more detailed evaluation is required, particularly to account for dynamic effects, large displacements or fatigue, advanced analysis software such as OrcaFlex is employed. Examples of such applications are discussed in later sections of this report.

By undertaking a holistic assessment of constraints and considering availability of installation equipment, a robust solution can be developed that aligns with project objectives as described in the Case Study provided.

Case Study

Background

This case study describes the engineering output of a shallow water 32in (813 mm) OD pipeline project in the Caspian Sea where conventional offshore installation techniques were neither feasible nor economically viable. In this instance, the selected concept was to construct the pipeline into string lengths onshore and assess various installation techniques from a technical and constructability perspective, considering a range of local constraints including:

- Reducing level of the Caspian Sea
- Limited onshore space for pipeline fabrication
- Ultra-shallow water (0 to 2 m) for approximately 35 km
- Offshore section approximately (> 2 m) 30 km long
- Caspian Sea land-locked resulting in limited equipment availability and logistical issues

The combination of these constraints resulting in a challenging project that required a hybrid solution consisting of the following:

- Onshore – consideration of pipeline construction and maximising string lengths whilst minimising installation pull loads
- Nearshore – consideration of pipeline installation through the 35 km ultra-shallow section and method of reducing installation loads whilst maximising pipeline string lengths
- Offshore – consideration of various installation methods ensuring compatibility with the onshore pipeline construction and launch through the nearshore section

Design Data and Analysis Inputs

Design Data.

Wind Data

For this analysis, the extreme wind speed associated with a 1-year return period was used to assess wind loading on sections of pipeline above the waterline. Specifically, the offshore 1-hour mean extreme wind speed measured at 10 m above mean sea level is applied within the standard formula for estimating wind speeds at different elevations.

$$U(z) = U(o) * \left[1 + C * \ln\left(\frac{z}{10}\right) \right]$$

$$C = 5.73 * 10^2 * (1 + (0.15 * U(o)^{0.5}))$$

$$U(o) = 1 \text{ hour mean wind speed} = 21.2 \text{ ms}^{-1}$$

Table 1—Wind velocity profile 1-year return period

Z (m)	C	U(o) (ms ⁻¹)	U(z) (ms ⁻¹)
5	0.12	21.2	19.5
6			19.9
7			20.3
8			20.7
9			20.9
10			21.2
11			21.4
12			21.7
13			21.9
14			22.0
15			22.2
16			22.4
17			22.5
18			22.7
19			22.8
20			22.9

Wave Data

Directional wave heights and associated wave periods corresponding to a 1-year return period were systematically summarised. This information provides a basis for evaluating environmental loading and operational limitations with respect to prevailing sea states from various directions.

Current Profile

An omni-directional current profile corresponding to a 10-year return period has been employed in the analysis. This long-term representation accounts for extreme current conditions irrespective of direction, ensuring a conservative basis for evaluating hydrodynamic loading and structural integrity.

Pipe Data

The line pipe data including key parameters such as pipe diameter, wall thickness, material grade and CRA mechanical properties are presented below:

Limiting Bending Strain for Installation Condition

The limiting integrity criteria for evaluating different installation techniques is the CRA wrinkling bending strain detailed below.

Analysis Inputs. In accordance with DNV-ST-F101, Table 2–4, the period from installation to pre-commissioning is classified as a temporary phase and assigned a Safety Class Low. As stipulated in Table 4–2 of the same standard, environmental load combinations involving 1-year and 10-year return periods for wave, current, and wind are permissible, provided the overall annual probability of occurrence does not exceed 10^{-2} .

Table 2—Wave extreme conditions 1-year return period

Wave Direction	Wave Height (m)	Wave Period (s)
North	1.30	4.08
North East	1.27	4.05
East	1.33	4.11
South East	1.37	4.15
South	1.45	4.21
South West	2.16	4.76
West	1.93	4.60
North West	1.32	4.10

Table 3—Omni-directional current profile 10-year return period

Depth Ratio	Current Velocity (m/s)
1.00	0.70
0.90	0.69
0.80	0.68
0.70	0.66
0.60	0.64
0.40	0.62
0.30	0.60
0.20	0.56
0.10	0.51
0.01	0.37

Table 4—Pipeline parameters

Parameter	Value	Units	Parameter	Value	Units
Outside Diameter	812.8	mm	CRA Type	Alloy 825	-
Wall thickness	22.23	mm	CRA Thickness	2.5	mm
SMYS	450	MPa	CRA Young's Modulus	197	GPa
SMTS	535	MPa	CRA Density	8440	kg/m ³
Steel Grade	API 5L X65	-	CRA SMYS	415	MPa
Young's Modulus	207	GPa	CRA SMTS	827	MPa
Steel Density	7850	kg/m ³	Concrete Weight Coating Thickness	60	mm
Poisson's ratio	0.3	-	Concrete Weight Coating Density	3040	kg/m ³

Table 5—Limiting bending strain

Steel Thickness (mm)	CRA Thickness (mm)	Limiting Bending Strain (%)	
		Girth Weld	Mid-Section
22.23	2.50	0.216	0.255

Given the anticipated dominance of hydrodynamic loading from waves and currents over wind forces during pipeline installation, particularly at the water surface, the following conservative approach was initially adopted:

- 1-year return period directional wave conditions
- 10-year return period omni-directional current conditions
- 1-year return period wind conditions

These combined inputs are used to assess the pipeline's strain response under environmental loading conditions as shown below.

Table 6—Environmental load case matrix

Load Case	Direction	Wave Height (m)	Wave Period (s)	Current Direction (°)	Wind Direction (°)
1	North	1.30	4.08	-26.15	-26.15
2	North East	1.27	4.05	18.85	18.85
3	East	1.33	4.11	63.85	63.85
4	South East	1.37	4.15	108.85	108.85
5	South	1.45	4.21	153.85	153.85
6	South West	2.16	4.76	198.85	198.85
7	West	1.93	4.60	243.85	243.85
8	North West	1.32	4.10	288.85	288.85

Onshore

All the offshore installation methodologies considered have similar onshore pipeline fabrication requirements. Prior to fabrication, assembly and launch of the pipeline, onshore site preparation typically includes:

- Identify suitable locations for onshore works
- Access consideration from road / sea
- Ground preparation requirements
- Material storage areas
- Fabrication facilities (workshop for pipeline and buoyancy fabrication)
- Pipe string storage area
- Pipeline assembly line
- Pipeline launch line

Pipe Fabrication Facilities. To support efficient pipeline fabrication, a covered fabrication shed was proposed at the inshore end of the launch site. This facility was designed to accommodate the full

sequence of fabrication activities, including pipeline fit-up, girth welding, non-destructive testing, and coating operations. These stations were arranged to enable the assembly of continuous pipeline strings from individual pipe joints.

Pipe strings are advanced along a series of conveyor rollers or rail tracks from the welding stations until the desired string length is achieved. Once completed, the pipe string is transferred and stored parallel to the firing line, either on concrete plinths (see Fig. 3) or timber skids. This relocation frees up the rollers or rail track for the fabrication of the next string.

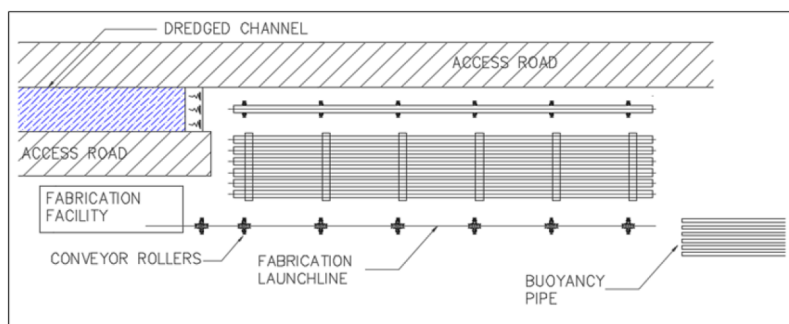


Figure 3—Typical Conveyor Roller Stringing Site Layout

Buoyancy can be achieved through various methods, including discrete modules or continuous polyethylene (PE) piping. For this project, continuous PE pipe was selected to offer greater flexibility in accommodating the proposed installation concepts.

When extended lengths of buoyancy pipe are required, welding operations may be conducted either within the main fabrication facility or at an alternative nearby site, depending on space availability. Upon completion, the buoyancy pipe strings should be stored adjacent to the main pipeline strings to ensure efficient site organisation and streamline subsequent handling operations.

Pipe String Launch Line. Two options were considered for fabrication and launch of the pipeline, one being the use of conveyor rollers and the other being the use of a railtrack.

Conveyor Roller Launch Line

Launching pipelines on conveyor rollers is standard industry practice for shore approaches and trenchless installations. This method reduces installation loads by an order of magnitude of ten compared to the pipelines self-weight, assuming a friction coefficient of 0.1.

Given the self-weight of the pipeline and the proposed use of excavators for launching through the nearshore section, string lengths were limited to approximately 250 m. This configuration resulted in an estimated pull load of around 200 kN, ensuring compatibility with the available equipment and maintaining safe operational limits.

Rail Track Launch Line

Utilising a rail track system enables significantly longer pipeline strings compared to conveyor rollers, primarily due to its substantially lower friction (assumed to be 0.01). The friction reduction is approximately an order of magnitude around ten times lower than that of conveyor rollers. However, this advantage comes with the trade-off of requiring a significantly larger laydown area for the launch line, due to the extended string lengths.

In line with the conveyor rollers concept, where the pipeline is launched using excavators and the installation load is limited to 200 kN, the string lengths were restricted to approximately 2,500 meters.

Nearshore

Shallow water along the 35 km nearshore section required constructing a causeway on either side of a dredged channel to facilitate pulling long pipeline strings offshore. Buoyancy was added to the strings before

pull-in from the onshore site, reducing submerged weight and keeping loads within the limits of land-based equipment even as additional sections were welded to extend the strings.

The maximum overall fabricated string lengths pulled towards the offshore section were jointly dictated by equipment capacities and results from the numerical modelling detailed later in the case study.

Offshore

At the offshore end of the nearshore section the installation sequence is transferred to an offshore operation. To provide flexibility and adaptability to the project, four installation methodologies were evaluated and considered for implementation.

On-Bottom Pull Method. During offshore pipeline installation via the On-Bottom Pull Method, the pipeline is laid directly on the seabed and pulled into position using a winch system mounted on a vessel, typically a flat-top anchored barge. The winch is connected to the leading end of the pipeline and to a reaction anchor through long lengths of steel wire rope, as illustrated schematically below.

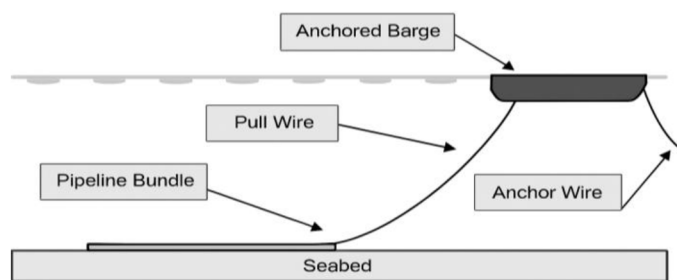


Figure 4—Schematic of On-Bottom Pull Arrangement

In this instance, the live wire was connected to the anchor wire, while the static wire was attached to the pipeline. Consequently, the barge advanced offshore, effectively pulling the pipeline along the seabed behind it. Due to the interaction with the seabed, an abrasion-resistant coating and a clear seabed is required.

Maximising the length of pipeline installed during each winching operation significantly reduces the number of AWTIs required, which is a primary output from the analytical assessment. This, in turn, delivers notable schedule and cost efficiencies, streamlining offshore operations and minimising vessel time and weather-related risk.

Analytical Assessment

As conventional On-Bottom Pull installations are designed to ensure minimal bending stresses are induced into the pipeline, numerical modelling to assess bending strain was not undertaken.

A sensitivity assessment was conducted to determine the maximum length of pipeline bundle that could be pulled across the seabed under varying buoyancy configurations, using derated linear winch capacities of 240 t and 640 t (from nominal 300 t and 800 t ratings) to account for offshore operational conditions and unforeseen loading factors. Results from the assessment are presented below based on pipe to seabed friction coefficient of 0.9.

Subject to verification of on-bottom stability requirements for the proposed installation window, an assessment outside the scope of this paper, the recommended buoyancy configuration for both linear winch systems consisted of two continuous (PE) pipes with an OD of 400 mm, resulting in optimal string lengths of 2.8 km and 7.5 km for the 300 t (2924 kN) and 800 t (7845 kN) winch systems, respectively.

Table 7—Pipeline Bundle Weight and Pull Force Assessment for On-Bottom Installation Method

Description	Unit	Buoyancy PE Pipe Configuration						
Buoyancy Pipe OD	mm	-	160	225	280	355	400	450
Number of Buoyancy Pipe	No.	-	2	2	2	2	2	2
Buoyancy Uplift	kg/m	-	18	35	54	87	110	139
Pipeline Weight Submerged, complete with Buoyancy	kg/m	311	276	241	203	137	91	33
Maximum Available Pull Force	tonne	240	240	240	240	240	240	240
Maximum Length of Pipeline for 300t system	m	858	967	1106	1314	1940	2919	8052
Maximum Available Pull Force	tonne	640	640	640	640	640	640	640
Maximum Length of Pipeline for 800t system	m	2288	2579	2949	3504	5174	7785	21472

Surface Tow and Float and Flood Method. The Surface Tow and Float and Flood Method allows the installation of pipelines in both shallow and deeper water. It offers the advantage that pipelines can be fabricated on land, launched, towed to the required location, and quickly and accurately laid on the seabed without the need for specialist installation vessels.

During installation, flooding of the buoyancy pipe (or product pipe) commences, allowing the bundle to take an S-shape to touch down. Once the bundle is installed and situated on the seabed the buoyancy pipe can be removed, recovered and surface towed back to the fabrication site for re-use.

The pipeline launch from the stringing site and subsequent pull through the dredged channel followed the procedure outlined in the nearshore section. Upon reaching the transition point to the offshore section, the pipeline, complete with buoyancy, will be surface towed to the designated installation location. Shallow draft vessels were selected for this operation to ensure accessibility and control to remain over the pre-dredged trench and away from the side cast dredged material.

This operation utilises a standard offshore tow spread, as illustrated below, ensuring controlled movement and stability during transit.

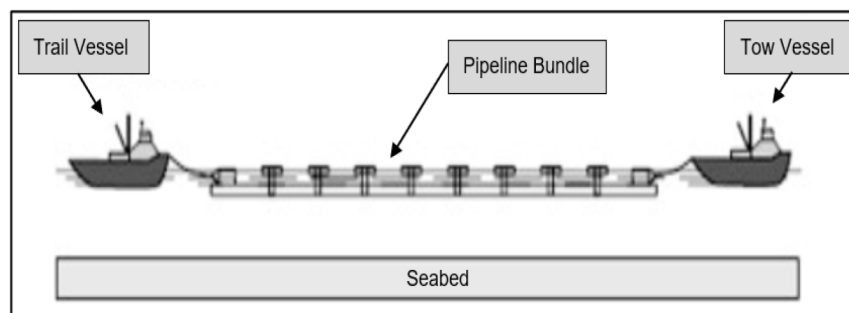


Figure 5—Schematic of Surface Tow Pull Arrangement

Buoyancy assessments determined that the optimal configuration for ballast and tow control was the use of two continuous PE pipes, each with an OD of 500 mm, or a single continuous PE pipe with a 710 mm OD. This solution provided the required buoyancy and stability to ensure controlled surface towing of the pipeline to the installation location.

Numerical Modelling

A strain-based assessment was undertaken to determine the maximum permissible string lengths during towing at a constant 0.5 m/s (1 knot) tow speed. A computational study was performed using Orcaflex considering pipeline lengths of 1 km, 7.5 km and 20 km, aimed at identifying the maximum installable pipe string length that remains within the CRA wrinkling strain threshold.

For the portion of the pipe that floats above the sea surface, wind loading was modeled with a vertical variation factor, accounting for height-dependent wind velocity near the waterline.

However, when applying the onerous environmental parameters, specifically the hydrodynamic loadings (wave and current) perpendicular to the pipeline, the results indicated that the permissible string lengths were not practicable as this would require a significant amount of AWTIs offshore making the operation logistically and economically unfeasible.

To ensure the analysis reflected a realistic constructability scenario and reduce lateral deflection, the assessment was re-run using estimated conditions representative of a benign environmental window. This approach aligns with standard industry practice when planning pipeline installations within a short-duration weather window, typically less than three days. By applying these more favorable parameters, the analysis aimed to evaluate the feasibility of towing operations under conditions that are both operationally achievable and logistically achievable.

The figure below presents results from an evaluation of maximum tension versus bending strain for three pipe string lengths. In each case, a single towing vessel was positioned at the leading end of the pipe string, while a rear vessel provided nominal holdback from the tail. The assessment was conducted under a benign environmental window, characterised by:

- Negligible wind and current conditions
- Wave height limited to 1 m, based on assumed vessel operability limitation
- Single Tow Tug/Vessel: 30 to 40 t bollard pull
- Single Trail Tug/Vessel: 10 to 20 t bollard pull

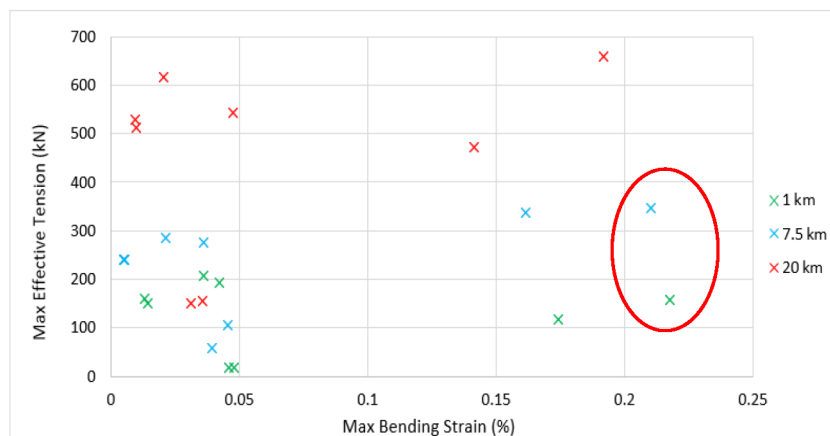


Figure 6—Effective Tension vs Bending Strain for Single Tow Vessel Case – Various String Lengths

As illustrated in the above graph, strain outliers in two towing simulations for Load Case 4 and 7 exceeded the wrinkling limit for the CRA. These exceedances were primarily attributed to elevated wave heights acting on individual pipe segments. To evaluate mitigation strategies, the affected cases were reanalysed using a dual vessel towing configuration over a 7.5 km pipe length. Although the strain results for the 20 km cases were within the CRA capacity, tow loads were considered too high and therefore discounted from further assessment.

To enhance realism, a ramping speed profile was applied to simulate initiation from a stationary state. However, dynamic deceleration scenarios, such as sudden stopping or gradual slowdown, were not considered to occur with normal operational conditions and are generally managed by:

- Emergency stopping protocols
- Controlled ramping down procedures

In this configuration, two towing vessels travel at a controlled speed of 0.5 m/s. One vessel is positioned at the leading end of the pipe string, while the second is strategically located mid-way along the string to move in unison, providing enhanced control over pipeline curvature and axial tension. A single trailing tug is stationed at the rear to offer nominal holdback, helping to stabilise the tail end and mitigate excessive strain.

In practice, multiple support vessels are typically deployed during towed pipeline installations. These vessels provide lateral restraint to manage curvature and can contribute additional tensile capacity when required. However, a detailed assessment of such multi-vessel configurations lies beyond the scope of this paper.

Typical surface tow spreads for this method include:

- Dual Tow Tug/Vessel: 30 to 40 t bollard pull
- Single Trail Tug/Vessel: 10 to 20 t bollard pull

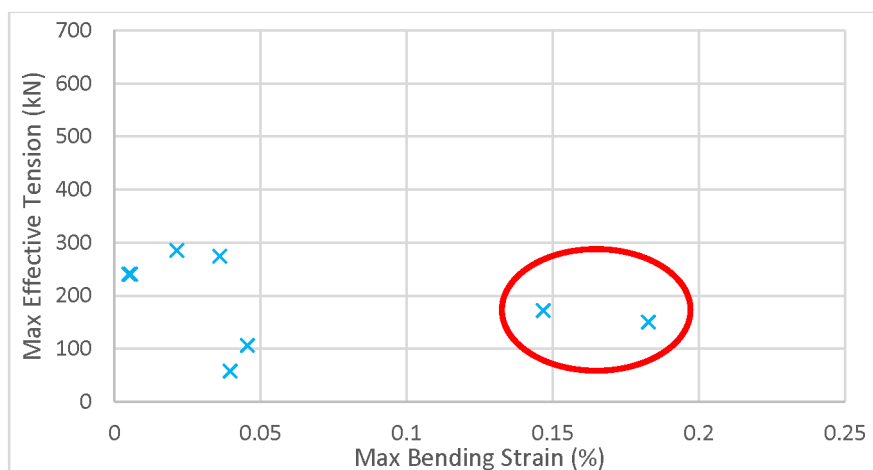


Figure 7—Effective Tension vs Bending Strain for Dual Vessel Case - 7.5 km Long String

As illustrated in the above graph, the simulation results for a dual vessel tow effectively reduces maximum strain levels and tensile loads experienced along the pipeline. The results from the two assessments are summarised in the table below.

These findings highlight the operational advantages of dual vessel towing when managing long, flexible multiphase pipelines in challenging offshore environments, particularly where stress and strain capacity is limited. The dual configuration enhances control over curvature and tension distribution, helping to mitigate peak strain and reduce the risk of structural instability.

Table 8—Surface tow sensitivity results for 7.5 km Long String

Load Case	Single Tow		Dual Tow	
	Strain (%)	Tension (kN)	Strain (%)	Tension (kN)
4	0.162	338	0.147	172
7	0.211	348	0.180	150

Off-Bottom Tow Method

The Off-Bottom Tow Method is an adaptation of the traditional Surface Tow and Float and Flood delivery system. This technique employs a combination of buoyancy and ballast chains to suspend the pipeline or bundle just above the seabed, allowing it to remain clear of direct contact while still maintaining stability and controlled seabed interaction through the ballast chain. This configuration provides enhanced tow control, reduces seabed disturbance, and mitigates hydrodynamic loading, particularly in shallow or environmentally sensitive areas.

The pipeline launch from the stringing site and subsequent pull through the dredged channel followed the procedure outlined in the nearshore section, except for the additional operation of adding ballast chain. The chain will be attached to the pipeline prior to launch and stowed clear of the rollers / railtrack.

Upon reaching the transition point to the offshore section, straps holding the ballast chain are released and the pipeline, complete with buoyancy and chain, is towed off-bottom to the designated installation location. Shallow draft vessels were selected for this operation to ensure accessibility in shallow water depths.

This operation will utilise a typical offshore tow spread, as illustrated below, ensuring controlled movement and stability during transit.

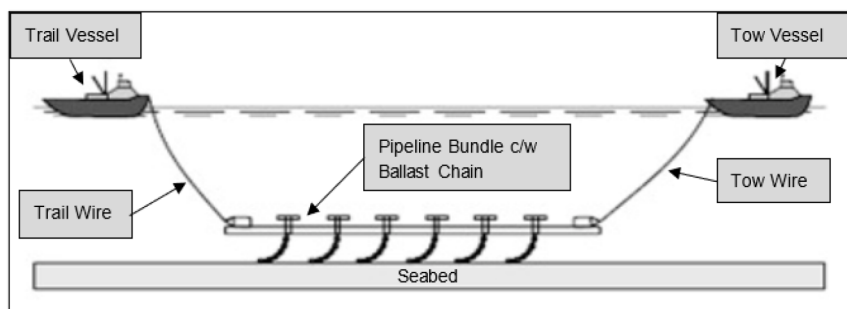


Figure 8—Schematic of Off-Bottom Tow Arrangement

Buoyancy assessments determined that the optimal pipeline configuration prior to chain attachment was either two continuous PE pipes, each with an OD of 500 mm, or a single continuous PE pipe with a 710 mm OD. These configurations were selected to achieve a target positive buoyancy of approximately 30 kg/m before chain ballast was added. To ensure negative buoyancy under static conditions, chain ballast was introduced to exert an additional downward force on the pipeline. This ballast increases the net submerged weight of the bundle to approximately 5 to 10 kg/m, ensuring the pipeline remains fully submerged and stable on the seabed.

However, during towing operations, the dynamics of the system change significantly. The forward motion of the tow vessel induces hydrodynamic drag on the chain ballast, which acts in the opposite direction of the towing force. This drag generates an upward component (lift) that partially counteracts the chain's weight, causing the pipeline bundle to rise through the water column. The system reaches equilibrium when the upward drag and lift forces balance the net downward force. Since drag force is proportional to the square of the towing speed, the vertical position of the bundle becomes highly sensitive to changes in velocity. As a result, careful control of towing speed and chain sizing is essential to maintain the desired off-bottom tow profile and avoid the ballast chains losing contact with the seabed.

The numerical modelling to assess CRA bending strain was based on the following input parameters:

- Chain ballast intervals – 10 m
- Chain length – 2.3 m
- Chain link diameter – 0.1 m

Numerical Modelling

A strain-based assessment was undertaken using Orcaflex for the 7.5 km string to provide results for the most onerous case. The following figure illustrates the pipeline geometry and chain ballast distribution as implemented in the model:

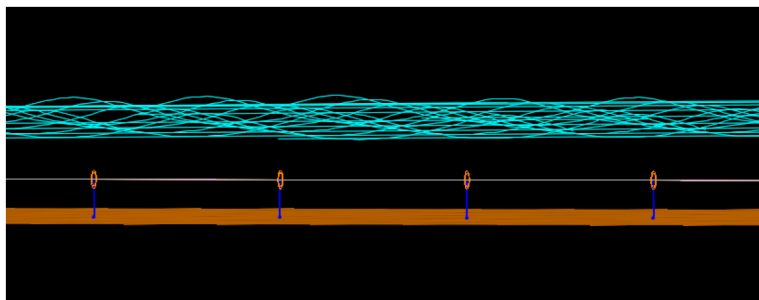


Figure 9—Schematic of Chain (navy) Distribution Along Pipeline (white)

The pipe string is suspended in the water column, therefore shielded from wind and surface current. It was found that due to the additional chain weight and friction with the seabed, the tension required to pull the pipe string was higher than the Surface Tow Method, with a mean value of 680 kN (69 t) during towing operation as shown below.

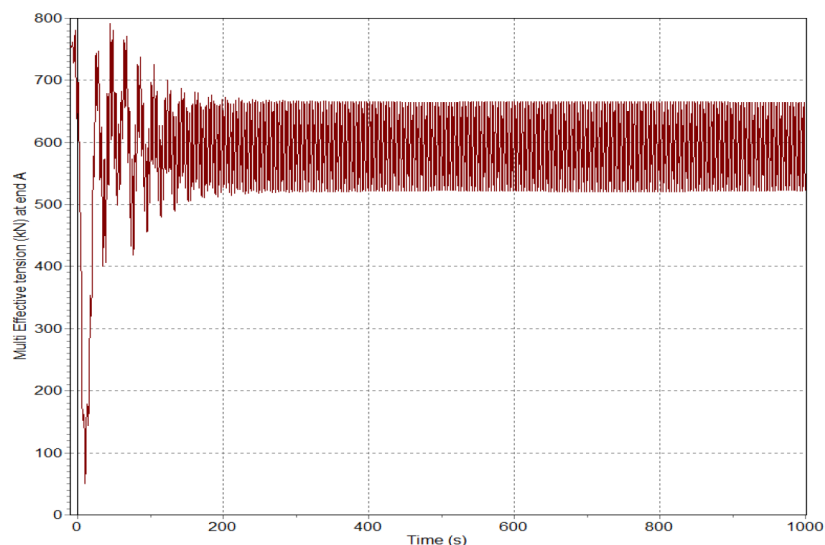


Figure 10—Off-Bottom Installation Wire Tension Profile – 7.5 km Long String

The maximum bending strain induced was calculated as 0.02% compared to 0.2% for the Surface Tow Method, primarily due to the pipeline being located lowered in the water column and partly shielded within a trench.

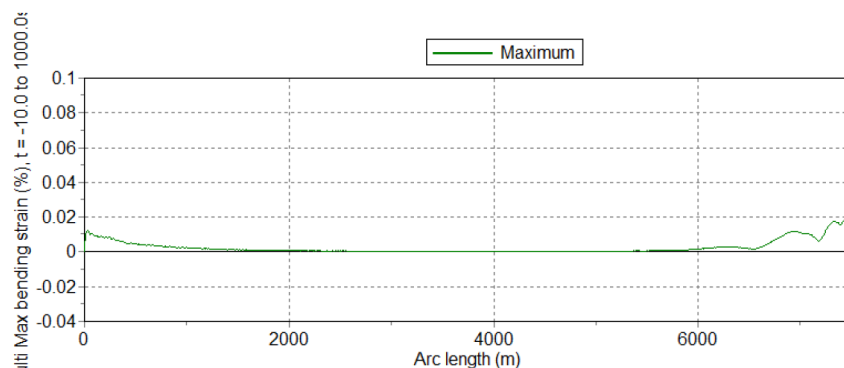


Figure 11—Bending Strain vs Arc Length – 7.5 km Long String

The results of the analysis demonstrate that the Off-Bottom Tow Method results in significantly lower bending strain compared to the conventional Surface Tow Method. However, this benefit comes with a trade-off; the axial tension required to pull the pipeline is notably higher due to the added weight of intermittently spaced chain ballast and increased frictional interaction with the seabed. For a 7.5 km pipe string, the tow tension was calculated to be approximately 680 kN (69 t), which exceeds the typical bollard pull capacity of shallow water installation vessels, which are generally limited to 294 kN (30 t). To ensure compatibility with conventional vessels assuming a linear relationship between pipeline length and required tow force, the recommended string length was calculated to be 3.3 km, based on the formula $(7.5 \text{ km} / 68 \text{ t}) \times 30 \text{ t} = 3.3 \text{ km}$.

Hybrid Method

The Hybrid Method combines the On-Bottom Pull and Surface Tow installation methodologies. Manipulating buoyancy positioning, the bundle string comprises floating buoyant sections and unbuoyant sections which sit directly on the seabed, as typically shown below:

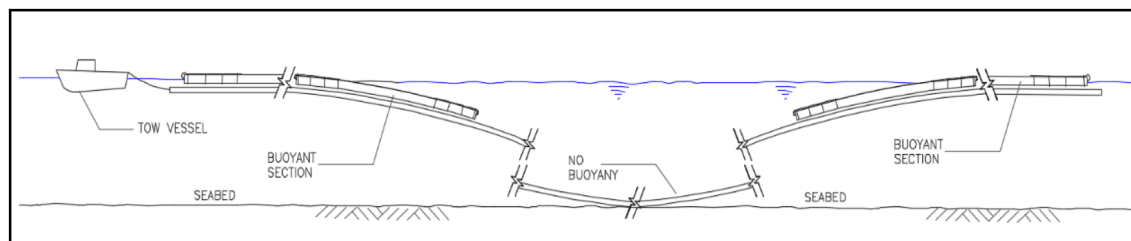


Figure 12—Schematic of Hybrid Method

The pipeline launch from the stringing site and its subsequent pull through the channel followed the procedure detailed in the nearshore section. Upon transition and handover to the offshore marine spread, selected buoyancy modules (pipe) are either flooded or removed to achieve a ‘sinusoidal’ configuration. In this setup, sections of the pipeline rest on the seabed, providing increased lateral restraint against hydrodynamic forces. This approach offers greater stability compared to the Surface Tow Method, which is inherently less stable during installation. However, the trade-off is the requirement for significantly higher tow forces due to the increased frictional resistance from seabed contact.

Buoyancy assessments determined that intermittent sections of two 900 mm OD buoyancy pipes provided the optimal distribution for minimising pipeline-induced stresses at the considered water depths. Unbuoyant sections were kept as short as possible to reduce installation loads. It should be noted that the ratio of buoyant to unbuoyant sections will vary in respect to increasing water depths and stability requirements, necessitating an iterative assessment which falls outside the scope of this paper.

Numerical Modelling

A strain-based assessment was undertaken using Orcaflex for a 7.5 km long string with alternating 800 m buoyancy and unbuoyant 160 m bare pipe sections, considering the most onerous hydrodynamic load case.

Due to the applied buoyancy distribution, the pipeline adopted a sinusoidal profile along its length. At the peak of the sag bend, the pipe made continuous contact with the seabed over a span of 15 m, generating sufficient frictional resistance to stabilise the pipe string.

An initial static assessment of bending strain revealed a consistent strain of 0.04% at the unbuoyant sections of the pipeline. In contrast, a higher strain of 0.1% was observed at the back-floating section near the tail end of the pipe string. Subsequent analysis indicated that introducing axial tension, either via a trailing tug or by adjusting the buoyancy distribution to maintain continuous seabed contact at the pipeline tail, effectively reduces this peak strain.

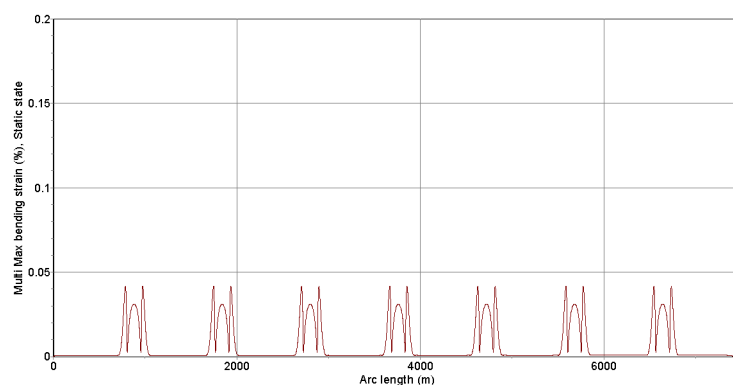


Figure 13—Strain vs Arc Length Static Assessment – 7.5 km Long String with 800 m / 150 m Buoyancy Distribution

Applying hydrodynamic loadings to the static pipeline configuration resulted in out-of-plane curvature, primarily driven by current and wind forces. These dynamic effects induced elevated bending strains that exceeded the wrinkling threshold for the CRA material. Analysis of dynamic response data revealed that up to 4 km of the pipe string experienced maximum bending strains of approximately 0.15%. Beyond this point, strain levels increased further, surpassing the CRA wrinkling limit of 0.216%.

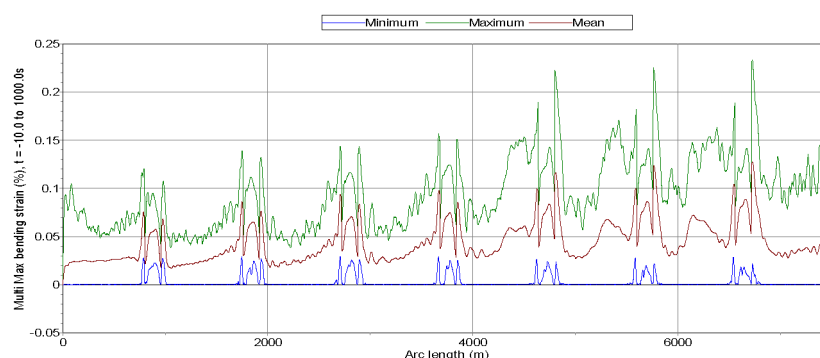


Figure 14—Strain vs Arc Length Dynamic Assessment – 7.5 km String Long with 800 m / 150 m Buoyancy Distribution

The following image provides a visual representation of the lateral deflection along the pipe string gradually increasing from the fixed end at the leading end of the pipeline.

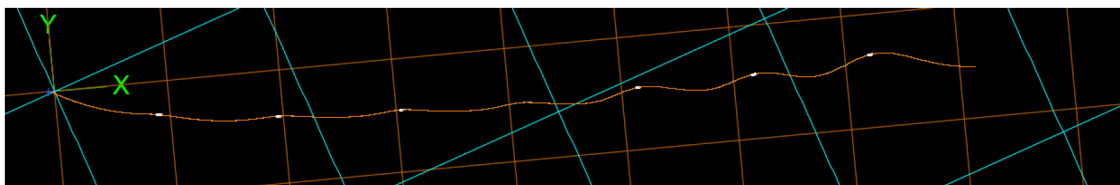


Figure 15—Pipe String Deflection Dynamic Assessment – 7.5 km Long String with 800 m / 150 m Buoyancy Distribution

Table 9—Results for Dynamic Assessment - 7.5 km String with 800 m / 150 m Buoyancy Distribution

Strain (%)	Tension (kN)
0.235	1106

As the calculated strain exceeded the allowable wrinkling threshold for the CRA, a sensitivity check was performed on shorter pipe strings with varying buoyancy configurations. The following figure presents the

strain distribution for a 5 km pipe string using the same buoyancy layout (800 m / 150 m), allowing for comparative assessment of strain behaviour under modified installation parameters.

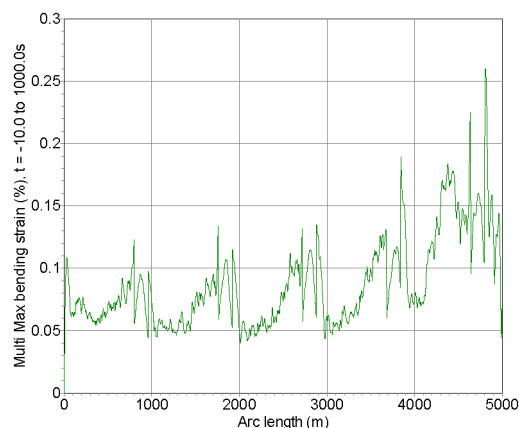


Figure 16—Strain vs Arc Length Dynamic Assessment – 5.0 km Long String with 800 m / 150 m Buoyancy Distribution

Table 10—Results for Dynamic Assessment – 5.0 km Long String with 800 m / 150 m Buoyancy Distribution

Strain (%)	Tension (kN)
0.260	738

Compared to the 7.5 km case with the same buoyancy distribution, the calculated strain increased toward the free end. This is likely attributable to the extended length of the floating pipe segment between the final touchdown point and the pipeline tail. Conversely, the installation tension was reduced by approximately 33%, which aligns with expectations due to the shorter overall pipe length and reduced number of sections in contact with the seabed.

A further iteration was undertaken to assess if the bending strain was reduced by increasing the number of sections in contact with the seabed. Keeping the string length at 5 km and setting the buoyancy distribution to 300 m / 150 m buoyant to unbuoyant, the strain distribution along the string is as follows:

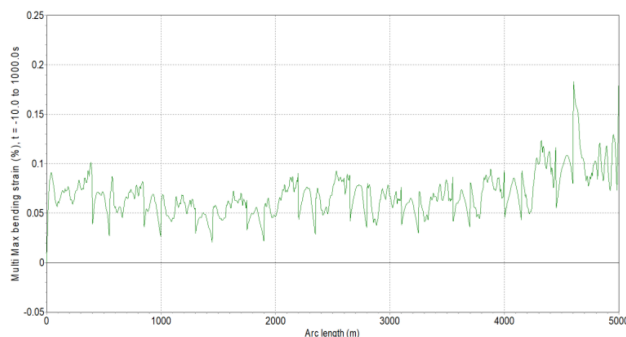


Figure 17—Strain vs Arc Length Dynamic Assessment – 5.0 km Long String with 300 m / 150 m Buoyancy Distribution

Table 11—Results for Dynamic Assessment – 5.0 km Long String with 300 m / 150 m Buoyancy Distribution

Strain (%)	Tension (kN)
0.193	955

The results demonstrate that lower bending strain can be achieved by increasing the number of pipeline sections in contact with the seabed. In this configuration, the maximum strain was approximately 0.19%,

representing 88% of the allowable wrinkling limit for the CRA. However, this reduction in strain comes at the cost of increased installation tension, calculated as 955 kN.

A sensitivity analysis was conducted for the dual tow method, which demonstrated negligible advantages compared to the single tow method. Consequently, it was excluded from further evaluation on the grounds of impracticality as connecting secondary tow vessels at intermittent floating sections was not considered feasible.

Summary, Discussion and Results

The following table summarises each installation method considered in terms of recommended length, buoyancy configuration, estimated installation force and calculated CRA bending strain. The table also provides high-level summary of the core offshore spread / equipment required for each option.

Table 12—Installation method summary and results

Methodology	Recommended Length (km)	Buoyancy / Ballast Configuration	Installation Pull / Tow Force Required (kN/t)	Calculated Bending Strain in CRA (%)	Core Offshore Spread / Equipment Required
On-Bottom Pull Method ^(a)	2.8 (300 t system) / 7.5 (800 t system)	Continuous Buoyancy Pipe: 2 No. 400mm OD	2357 / 240 (300 t system) 6276 / 640 (800 t system)	-	Flat top barge (or similar) complete with mooring winches, linear winch and associated equipment Reels and reel winders complete with pull wire AWTI barge complete with mooring winches and davit system / craneage 20 to 30 t Multicat (anchor handling) or similar Support vessels
Surface Tow and Float Flood Method	7.5	Continuous Buoyancy Pipe: 2 No. 500 mm OD, or 1 No. 710 mm OD	170 / 17 (Dual Tow) ^(b)	0.180 (Dual Tow) ^(b)	30 t to 40 t bollard pull tow tug / vessel 10 t to 20 t bollard pull trail tug / vessel AWTI barge complete with mooring winches and davit system / craneage Several support vessels
Off-Bottom Tow Method	3.3	Continuous Buoyancy Pipe: 2 No. 500 mm OD, or 1 No. 710 mm OD Chain Ballast: Spacing – 10 m Length – 2.3 m OD – 0.1 m	294 / 30	0.02	30 t to 40 t bollard pull tow tug / vessel 10 t to 20 t bollard pull trail tug / vessel AWTI barge complete with mooring winches and davit system / craneage Several support vessels
Hybrid Method ^(c)	5.0	Intermittent Buoyancy Pipe: 2 No. 900 mm OD 300 m buoyant / 150 m unbuoyant	955 / 95	0.193	Flat top barge (or similar) complete with mooring winches linear, winch and associated equipment Reels and reel winders complete with pull wire AWTI barge complete with mooring winches and davit system / craneage 20 to 30 t Multicat (anchor handling) or similar Several support vessels

Notes:

^(a) Recommended length will be dictated by stability requirements

^(b) Maximum values from load cases considered

^(c) Buoyancy distribution optimised for water depths in the region of 2 to 3 m

From a technical feasibility standpoint, all proposed installation methodologies are considered suitable for addressing the unique site-specific constraints present in the northern region of the Caspian Sea. These constraints, including shallow water depths, soft seabed conditions, and seasonal environmental variations, can be effectively managed through tailored engineering solutions across the available methods.

However, from a constructability perspective, the Hybrid Method poses the most significant challenges. This is largely due to its dependence on intricate buoyancy control systems, which must be carefully assessed in relation to variable water depths. In a geographical setting characterised significant tidal ranges, these complexities are further exacerbated. During low tide conditions, increased pipe contact with the seabed can lead to elevated installation loads, potentially limiting installation windows. Conversely, high tide cycles may reduce seabed contact length, altering the designed stability case. These fluctuations introduce considerable uncertainty into installation planning and execution, demanding highly responsive and adaptive construction strategies.

Each concept will require support from shallow-draft vessels to perform typical marine operations such as wire transfer, hook-up, and anchor handling. These activities demand careful planning and close coordination with dredging strategies, particularly where access channels are required to accommodate vessel entry and manoeuvring in ultra-shallow waters.

The case study presented demonstrates innovative alternatives when conventional offshore techniques prove unfeasible or unsuitable. The distinctive topography and bathymetric conditions of the northern Caspian Sea enabled a unique solution by utilising land-based equipment to install sections of the pipeline, delivering notable commercial advantages to the project.

Onshore infrastructure included dedicated pipeline fabrication facilities and a conventional launch setup, allowing the pipeline string to be pulled from the stringing site into a pre-dredged channel. This approach not only optimised operational efficiency but also minimised reliance on specialised offshore vessels, reducing overall project costs and logistical complexity.

Side-casting excavated material to construct a continuous access causeway not only facilitates logistical movement but also serves a dual function by providing a protective barrier for the floating pipeline bundle. This configuration effectively mitigates exposure to hydrodynamic forces and wind-induced loadings, thereby reducing weather-related risks. As a result, it provides greater flexibility by broadening the range of viable weather windows for planning offshore works.

The following sub-sections offer a high-level summary of the primary constructability advantages and limitations associated with each installation methodology.

On-Bottom Pull Method

Advantages:

- Minimal installation-induced stresses during installation
- Pipeline located on the seabed mitigating risks typically associated with surface-towed solutions
- Reduced number of AWTIs, due to extended bundle string lengths, subject to stability assessments
- Less susceptible to adverse environmental conditions than traditional towed methods

Limitations:

- Requires specialist equipment, including linear winch spreads and pull wires
- Longer installation duration compared to conventional towed concepts
- Risk of concrete weight coating abrasion
- Challenges associated with anchor handling in shallow water

Surface Tow and Float and Flood Method

Advantages:

- Shorter installation duration compared to On-Bottom and Hybrid methods
- Continuous visibility of the bundle string during installation, enhancing monitoring and control
- Utilises non-specialist equipment, such as shallow-draft tugs and multicats vessels
- No abrasion concern as pipeline bundle is floating throughout the installation
- Pipeline bundle can be constructed remotely and towed to the installation location

Limitations:

- Bundle string length constrained by environmental conditions, bending strain limits, and tow force requirements (longer strings necessitate additional support vessels)
- High susceptibility to environmental conditions, which can introduce schedule risks and operational delays
- Flooding and venting of buoyancy systems must be performed while the bundle is floating, increasing operational complexity and risk
- Greater number of support vessels required compared to alternative installation techniques

Off-Bottom Tow Method

Advantages:

- Shorter installation duration compared to On-Bottom Pull and Hybrid methods
- Utilises non-specialist equipment, such as shallow-draft tugs and multicats vessels
- No abrasion concerns, as the pipeline bundle remains suspended in the water column throughout installation.
- Pipeline bundle can be constructed remotely and towed to the installation location
- Minimal installation-induced stresses during installation
- Less susceptible to adverse environmental conditions than traditional towed methods
- Lower installation loads compared to the On-Bottom Pull method

Limitations:

- Greater susceptibility to environmental conditions than On-Bottom Pull and Hybrid methods
- Additional offshore operations required for ballast chain recovery
- Higher pull loads compared to surface tow methods

Hybrid Method

Advantages:

- Pipeline partially located on the seabed which increases stability compared with surface towed solutions
- Reduced sensitivity to environmental conditions than towed solutions

Limitations:

- Requires specialist equipment, such as drum winches, linear winch spreads, and pull wires
- Longer installation duration compared to conventional towed concepts
- Floating sections are more vulnerable to environmental conditions than the On-Bottom Pull Method
- Bending stresses induced during towing for extended periods, due to the sinusoidal geometry of the bundle
- Dependency on intricate buoyancy control systems in respect to water depth

Conclusion

The case study highlights the need to tailor pipeline installation solutions to site-specific constraints and commercial drivers. Bathymetric variation, geotechnical conditions, environmental sensitivities, and logistical limits require a flexible, adaptive engineering approach. By applying customised methods—such as land-based equipment, hybrid techniques, or modified access strategies—each project should be evaluated individually to ensure technical feasibility and cost-effectiveness.

The case study highlighted the benefits of maximising pipeline string lengths, primarily through the reduction of offshore operations, particularly those involving AWTIs. Further optimisation may be achieved through advanced modelling of support vessel configurations along the floating pipeline length. Strategically positioning these vessels can enhance stability, reduce stress / strain concentrations, and facilitate safer handling during installation. Concrete weight coating, commonly applied to shallow water pipelines for stability and safety, introduces added rigidity that can significantly affect the pipeline's response to hydrodynamic loading and installation-induced strains. In this instance, the combined geometry of the pipeline and coating did not adversely impact CRA induced strains; however, such effects should be evaluated on a case-by-case basis to ensure integrity under varying conditions.

The study concluded that the available metocean data was excessively conservative and impractical for offshore operational planning. As a result, benign environmental conditions were adopted, with contingency measures in place to ensure pipeline safety in the event of adverse weather, an approach widely accepted for short-duration offshore activities. However, for operations exceeding three days, regulatory standards such as *DNV-RP-F109* and *DNV-ST-F101* require environmental simulations to be based on return period conditions, typically the seasonal 10-year wave combined with the 1-year current (or vice versa), rather than on assumed benign conditions. This shift may yield less favourable outcomes when evaluating pipeline stability and planning offshore works.

To ensure a representative and robust evaluation, it is recommended that shallow-water hydrodynamic effects, including shoaling, wave breaking, and refraction be explicitly integrated into simulation models. These phenomena significantly influence nearshore wave behaviour and load distribution. Their inclusion is essential to avoid overly conservative designs, which can lead to unnecessarily heavy pipelines or impractically narrow installation windows.

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Nomenclature

- o = wind speed reference height, m
- z = height above sea level, m
- C = atmospheric constant

U = wind speed, ms⁻¹

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